MILITARY SPECIFICATION

AIR REFUELING STORE SYSTEM; D-704

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. Scope

1.1 Scope: This specification establishes the requirements for the manufacture, inspection and testing of an air refueling store and its associated ground handling equipment and special tools. In the event of a conflict between this specification and the drawings furnished by the Government this specification shall have precedence.

1.2 Classification: The equipment covered by this specification shall be of one type designated as Air Refueling Store System: D-704. The store system shall be comprised of the following parts, as specified. (see 6.2):

Part I - Air refueling store and console
Part II - Shipping container and dolly
Part III - Special support equipment

2. APPLICABLE DOCUMENTS

2.1 General: The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-J-5624	Jet Fuel, Grades JP-3, JP-4 and JP-5
MIL-A-8591	Airborne Stores and Associated Suspension Equipment, General Design Criteria for
MIL-Q-9858	Quality Control System Requirements
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

Douglas Aircraft Company, Inc.

DPS2.484-1	Douglas In-Flight Fueling Store Tank

FSC 1680

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DRAWINGS:

U. S. Air Force

44M15935	Vibrator and Rocker Assembly Jack

Douglas Aircraft Company, Inc.

4546967-501	Sleeve-Reel In Snubbing Valve
5547000-523	Store Assy-Inflight Fueling Store
5814092	Rework-Lighted Drogue and Coupling Assy, Air Refueling
7552668-505	Container Specification - Inflight Fueling Store
7552770	Special Tools and Ground Handling Equipment - Inflight Fueling Store
7555940-503	Procedure - Adjustment, Test and Check Out, In-Flight Fueling Store

Harbor Boat Building Company

1504	Container Assy. - Shipping and Storage, In-Flight Fueling Store

PUBLICATIONS:

Air Force - Navy Aeronautical Bulletin

ANA-143	Specifications and Standards; use of

Bureau of Naval Weapons

NAVWEPS 03-100-502	Illustrated Parts Breakdown, In-Flight Fueling Store

(Copies of this specification and applicable specifications, standards, drawings and publications may be obtained from the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania (Attention: Code CDS)

3. REQUIREMENTS

3.1 Parts and Materials: Parts and materials used in the manufacture of the store system shall conform to the applicable specifications as specified herein, or as listed on the applicable drawings.

3.1.1 Selection of Specifications and Standards: Specifications and standards for all materials, processes, parts and equipment which are not specifically designated herein and which are necessary for the execution of this contract shall be selected in accordance with ANA Bulletin 143.

3.1.2 Standard Parts: Where applicable, component utility parts such as bolts, nuts, rivets, washers and connectors shall conform to AN standards or shall be interchangeable functionally and dimensionally with, or replaceable by, MS, AN or JAN standard parts without modification and shall be designated by their standard parts number.

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3.2 Design and Construction

3.2.1 Conformance to Drawings:

3.2.1.1 Part I: The air refueling store and console shall conform to Douglas Aircraft Company, Inc. drawing 5547000-523 except as follows:

(a) A high level float switch shall be incorporated in accordance with 3.7.7

(b) Drogue lighting shall be incorporated in accordance with 3.11.3

(c) The hole in the sleeve of the hose reel in snubbing valve, Douglas Aircraft Company, Inc. drawing 4546967-501, shall be enlarged from $\frac{.055}{.057}$ diameter to $\frac{.068}{.069}$ diameter.

(d) The 20 inch spaced three point suspension provisions and lugs therefor shall be deleted.

3.2.1.2 Part II: The shipping container and dolly shall conform to Douglas Aircraft Company, Inc. specification drawing 7552668-505 and to Harbor Boat Building Company drawing 1504.

3.2.1.3 Part III: The special support equipment shall conform to Douglas Aircraft Company, Inc. drawings listed on Douglas Aircraft Company, Inc. drawing 7552770 except that the shipping container listed on drawing 7552770 shall conform to Part II of this specification and is not included under Part III.

3.2.1.4 Changes: With the exception of changes (a),(b),(c) and (d) of 3.2.1.1, no departure from Douglas Aircraft Company, Inc. drawings and subsidiary documents shall be made without prior written approval by the Bureau of Naval Weapons.

3.2.2 Interchangeability:

3.2.2.1 Part I: All parts having the same part number shall be interchangeable with each other. All equipment, assemblies, structures, components and detail parts thereof equivalent to a given part of the store as illustrated and listed in NAVWEPS 03-100-502 Illustrated Parts Breakdown shall be directly and completely interchangeable with respect to installation and performance in stores manufactured by all contractors.

3.2.2.2 Part II: All parts having the same part number shall be interchangeable with each other. All equipment, assemblies, structures, components and detail parts thereof equivalent to a given part of the part number 1504-505 shipping container and dolly as illustrated and listed on Harbor Boat Building Company drawing 1504 shall be directly and completely interchangeable with respect to installation and performance in shipping containers and dollies manufactured by all contractors.

3.2.2.3 Part III: Support equipment, test equipment and tools shall be common to and suitable for the D-704 air refueling store systems manufactured by all contractors.

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3.2.2.4 Publications: Technical publications shall be common to and suitable for use with D-704 air refueling store systems manufactured by all contractors.

3.3 Performance: The store shall operate properly and shall be suitable for air refueling under any combination of the following conditions:

(a) Airspeed: From 200 knots calibrated airspeed up to and including 300 knots calibrated airspeed but not to exceed 0.90 Mach number.

(b) Engaging Speed: Receiver aircraft engages the tanker at a relative speed up to and including 8 feet per second.

(c) Altitude: From sea level up to 35,000 feet.

(d) Temperature: Ambient temperatures from minus 65 degrees Fahrenheit up to plus 100 degrees Fahrenheit

3.4 Store Characteristics:

3.4.1 Electrical Power: The store shall utilize tanker aircraft electrical power for control of all store operations. The store shall be operable from 28 volt D. C. with no A. C. supply and shall also be operable from a combination of 115 volt 400 cycle A. C. and 28 volt D. C. The maximum electrical power required when operating entirely from a 28 volt D. C. power source shall not exceed 225 watts.

3.4.2 Weight: The dry weight of the store shall not exceed 725 pounds. This weight includes the complete store with hydraulic oil plus two support lugs but does not include the console.

3.4.3 Center of Gravity: The store center of gravity with the store empty shall be located at station 117.0 plus 0, minus 2.0. The store center of gravity when fueled with 300 gallons of JP-4 fuel, or JP-5 fuel, shall be located at station 92.0 plus 2.0, minus 2.0.

3.5 Tank:

3.5.1 Capacity: Useable fuel capacity shall not be less than 300 gallons.

3.5.2 Pressure: The tank section of the store shall withstand the following pressures:

(a) Operating: 50 PSIG
(b) Proof: 57 1/2 PSIG
(c) Ultimate: 75 PSIG

3.6 Structural Loads: The store shall withstand the following conditions:

(a) The loads of Specification MIL-A-8591. Arrested landings with the store 25 percent full of fuel. Non-arrested landings with the store full of fuel.

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(b) 600 knots calibrated airspeed but not to exceed 1.0 Mach number with the drogue retracted and the ram-air turbine feathered.

(c) Hose limit load of 2000 pounds acting in a horizontal plane at an angle of 18 1/2 degrees from the center of the level wind guide.

(d) One ejection separation from the aircraft with an applied force of 25,000 pounds.

3.7 Fuel System:

3.7.1 Fuels: The system shall be suitable for fuels conforming to Specification MIL-J-5624, Grades JP-4 and JP-5.

3.7.2 Design Pressure: The fuel transfer system components subjected to transfer pump pressure shall be suitable for the following pressures: 120 PSIG operating, 180 PSIG proof and 360 PSIG burst.

3.7.3 Transfer Pressure: The fuel pressure at the reception coupling shall be between 35 and 50 PSIG, when pumping fuel at 0 to 200 gallons per minute.

3.7.4 Surge: Surge pressure shall not exceed the proof pressure specified in 3.7.2 under normal operating conditions.

3.7.5 Transfer Rate: The store shall be capable of transfering 200 gallons per minute of JP-4 fuel with the fuel temperature at 70 degrees F and 35 PSIG pressure at the reception coupling.

3.7.6 Dumping: The store shall be capable of dumping a minimum of 75 percent of the fuel carried in the store within three (3) minutes.

3.7.7 High Level Float Switch: A single high level float switch shall be incorporated in the store. Its function is to provide a signal to control operation of a valve or pump in the tanker aircraft. This feature will permit pressure fueling of the store via the tanker aircraft pressure fueling system. The switch shall be installed adjacent to the level control pilot, vent and relief valve. The switch shall be located so that the shutoff level will be below the point of closing the vent and will shut off the fueling at the same level as during ground pressure fueling through the store's pressure fueling receptacle. Details of the high level switch float switch installation shall be designed by the contractor. There are enough spare pins in the electrical connector of the store to accommodate the switch wiring. Interchangeability with the P/N 5547000-523 store shall be maintained.

3.8 Hydraulic System:

3.8.1 Fluid: The store shall use hydraulic fluid conforming to Specification MIL-H-5606.

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3.8.2 Temperature: The hydraulic fluid temperature shall not exceed 225 degrees F under the most severe operating conditions. A temperature of 200 degrees F shall not be exceeded during 90 percent of the system operating time.

3.9 Hose Reel:

3.9.1 Time: Time for extension and time for retraction to and from the full trail position shall not exceed 15 seconds.

3.9.2 Response: The reel shall respond and accelerate at a rate fast enough to prevent looping or oscillation of the hose under conditions of 3.3.

3.10 Hose Cutter: The hose cutter shall operate satisfactorily from minus 65 degrees F to plus 150 degrees F.

3.11 Drogue:

3.11.1 Stability: The drogue shall prevent oscillation of the hose during free trail. The drogue shall be stable in flight at all speeds and altitudes of the refueling operation.

3.11.2 Drag: The drag of the drogue shall be related to the tanker airspeed as $D = 1.85\ q$, where "q" is the aerodynamic pressure.

3.11.3 Drogue Lighting: A wind-driven generator night lighting system for the drogue shall be provided. The drogue lighting system and its installation shall conform to Douglas Aircraft Company, Inc. drawing 5814092. All parts shall be contractor furnished.

3.12 Identification of Product: Nameplates and identification plates shall be similar to those shown on drawings specified in 3.2.1 except as follows:

(a) The identification nomenclature applied to the store shall be:

MODEL D-704 AIR REFUELING STORE

(b) The store part number shall be included .

(c) The applicable contract number shall be indicated.

(d) The name of the manufacturer shall be included.

3.13 Engineering Data: The Government will furnish to the contractor a set of Douglas Aircraft Company drawings covering the P/N 5547000-523 store. The contractor shall redraw only those drawings which require changes to delineate the store as delivered by the contractor. The contractor shall prepare and furnish a complete assembly breakdown and a numerical list of all drawings for the store as delivered. Drawings and lists prepared by the contractor shall be in accordance with Specification MIL-D-70327 as applicable. Drawing numbers shall be assigned in conformance with the contractor's drawing numbering system. One reproducible and three prints of drawings and lists shall be furnished . The contractor shall also prepare and submit an engineering report completely describing and defining all aspects in which the store differs from the P/N 5547000-523 store.

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3.14 Workmanship: Workmanship shall be consistent with requirements of the intended use of the equipment and shall conform to high quality aircraft practice.

3.15 Quality Control: The contractor shall provide and maintain a quality control system that complies with the requirements of Specification MIL-Q-9858. Notwithstanding this requirement, the contractor is in no way relieved of the final responsibility to furnish supplies or services as specified herein.

4. QUALITY ASSURANCE PROVISIONS:

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of Tests:

4.2.1 Preproduction Tests: Preproduction tests are those tests conducted on stores and containers representative of the production articles to be supplied under the contract for the purpose of determining that the stores and containers meet all the requirements of the applicable specifications.

4.2.2 Acceptance Tests: Acceptance tests are those tests accomplished on stores and containers submitted for acceptance under the contract and shall consist of the tests specified herein.

4.2.3 Component Tests: Component tests are those tests conducted on equipment covered by contractor or Douglas Aircraft Company, Inc. procurement specification.

4.3 Tests Required: The following tests are required:

(1) Part I — Air refueling store and console:

Preproduction tests	4.4
Acceptance tests	4.5
Component tests	4.6

(2) Part II — Shipping containers and dolly:
Tests in accordance with sections 5 and 7 of Douglas Aircraft Company, Inc. Specification Drawing 7552668

(3) Part III — Special tools:
Visual examination to confirm adherence to Douglas Aircraft Company, Inc. drawings. A functional test shall be performed on one electrical tester.

4.4 Preproduction Tests:

4.4.1 Test Specimens: The first three (3) stores fabricated by quantity production methods and tools shall be allocated to preproduction testing. No additional stores shall be delivered prior to preproduction acceptance.

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4.4.2 Scope of Tests: The preproduction stores shall be subjected to the following tests in the listed sequence:

Store No. 1:

(1) Examination of product 4.7.1
(2) Weight and center of gravity 4.7.5.1
(3) Leakage 4.7.4.2
(4) Proof pressure 4.7.4.3
(5) Slosh-vibration 4.7.5
(6) Hose load 4.7.7
(7) Accelerated load 4.7.6
(8) Ultimate pressure 4.7.4.4

Store No. 2:

(1) Acceptance 4.5
(2) Hose reel response 4.7.8
(3) Store endurance 4.7.9
(4) Hose cutter test 4.7.10

Store No. 3:

(1) Acceptance 4.5
(2) Hose load to 2000 pounds maximum 4.7.7
(3) Naval service suitability 4.7.11

4.4.3 Preproduction Acceptance: Defects or failures evidenced as a result of tests shall be reported to the Bureau of Naval Weapons with recommendations for correction of the defect. Upon approval by the Bureau of Naval Weapons the test article shall be modified and retested. When all tests have been satisfactorily completed a report of the test program shall be submitted to the Bureau of Naval Weapons for acceptance.

4.4.4 Disposition of Preproduction Stores: Store No. 1 used for structural tests will not be suitable for service use and shall be marked as unsuitable for flight. It shall be retained by the contractor until the contract has been completed at which time it may be scrapped. Stores No. 2 and 3 shall be inspected, repaired as necessary and refurbished by the contractor after preproduction tests. The part no. 2672493 plate, part no. 4672372 knife and part no. 4674605 plug shall be replaced in the hose cutter of Store No. 2 and in Store No. 3 if its hose cutter has been fired. The Stores No. 2 and 3 shall be resubjected to acceptance tests before delivery.

4.5 Acceptance Tests:

4.5.1 Test Setup: The test setup shall include the following equipment:

(1) A production test wind tunnel capable of producing a 200 knot (plus 10, minus 0 knots) airflow over the store ram air turbine and suitable for ground-operation tests of the store.

(2) A device to provide and control hose tension for ground-operation tests.

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4.5.1 Scope of Tests: Each production store shall be subjected to the following tests:

(1) Examination of product 4.7.1
(2) Pressure 4.7.2
(3) Ground operation 4.7.3

4.5.2 Rejections: The failure of any store, assembly, sub-assembly or piece to pass the acceptance tests required by this specification shall cause its tentative rejection, subject to retest as hereinafter specified.

4.5.3 Retest and Reinspection: Rejected stores, assemblies, sub-assemblies or pieces may, upon request of the contractor to the cognizant Government Inspector, be resubmitted for inspection and test provided a statement showing the corrective measures taken is furnished to the Inspector at the time of resubmission. Failure to pass inspection and test when resubmitted shall be cause for rejection of the unit tested.

4.6 Component Tests: Component tests shall be conducted in accordance with the requirements of the applicable component procurement specification.

4.7 Test Methods:

4.7.1 Examination of Product: The store shall be inspected to insure that it has been properly assembled, that the wiring has no shorts, grounds or discontinuities and that the workmanship and finish are in accordance with this specification.

4.7.2 Pressure: The pressure test of the fuel tank shall conform to Douglas Aircraft Company, Inc. Process Standard DPS2.484-1.

4.7.3 Ground Operation: The store shall be tested in accordance with all of the test and checkout requirements of Sections II and III of Douglas Aircraft Company Handbook 7555940-503.

4.7.4 Qualification Pressure Tests: The purpose of these tests is to demonstrate that the store withstands internal pressures without leakage or failure both before and after being subjected to the other structural tests.

4.7.4.1 Test Setup: The store shall be mounted in a test fixture which supports the store on an Aero 7A rack to simulate aircraft installation. For this test all openings of the store but one are plugged and pressure is applied at the one opening. The nose and tail cone of the store are removed during this test. For the leakage test, 4.7.4.2, a mercury manometer shall be used to measure the internal pressure. For the proof test, 4.7.4.3 and the ultimate test, 4.7.4.4, a calibrated pressure gage shall be used to measure the internal pressure.

4.7.4.2 Leakage Test Procedure: The store shall be filled with a water solution having a surface tension equivalent to JP-4 fuel. Air pressure shall be applied to establish 15 PSIG internal pressure in the tank. Then the air supply shall be disconnected and the manometer shall be watched for any indication of pressure drop. Also the store shall be inspected for any sign of leakage. No pressure drop or leakage shall occur within 15 minutes of disconnecting the air supply.

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4.7.4.3 Proof Pressure Procedure: The store shall be filled with water solution having a surface tension equivalent to JP-4 fuel. Air pressure shall be applied to establish 57 1/2 PSIG internal pressure in the tank. Then the air supply shall be disconnected and the pressure gage shall be watched for any indication of pressure drop. Also the store shall be inspected for any sign of leakage. No pressure drop or leakage shall occur within one minute of disconnecting the air supply. The store shall be thoroughly inspected and no damage or deformation shall be evident.

4.7.4.4 Ultimate Pressure Procedure: The proof pressure test procedure, 4.7.4.3, shall be followed. After the test at 57 1/2 PSIG for one minute the air pressure shall be increased to 75 PSIG. Then the air supply shall be disconnected and the pressure gage shall be watched for any indication of pressure drop. Also the store shall be inspected for any sign of leakage. No pressure drop or leakage shall occur within 3 minutes of disconnecting the air supply. The store shall be disassembled and inspected. There shall be no structure failure such as severely deformed, cracked, loose or broken parts or members.

4.7.5 Slosh-Vibration Tests: The purpose of these tests is to demonstrate that the store withstands slosh and vibration loads.

4.7.5.1 Test Specimen: The store shall be a complete production article. The store shall be weighed and its center of gravity measured to determine compliance with 3.4.2 and 3.4.3.

4.7.5.2 Test Setup: The store shall be mounted in a test fixture which supports the store on an Aero 7A rack to simulate aircraft installation. The fixture shall be mounted on the table of a slosh-vibrator machine which conforms to USAF Drawing 44M15935 or equivalent design which has previously been acceptable to a Government Procuring Service. The store shall be instrumented to record the amplitude versus slosh-angle of the store at stations 24, 107, 140 and 200.

4.7.5.3 Test Procedure: The store shall be filled to two-thirds of its capacity with a water solution having a surface tension equivalent to JP-4 fuel. The machine shall be adjusted to produce a total amplitude of 0.032 plus 0.010, minus 0.000 inch at the top of the store at station 107 throughout the full slosh angle with a vibrational frequency of 2000 plus 0, minus 60 cycles per minute. The slosh rocking angle shall be 30 degrees total, approximately 15 degrees on either side of the horizontal position and the frequency shall be 18 cycles per minute. The store shall be subjected to simultaneous slosh and vibration for 12 1/2 hours in roll and 12 1/2 hours in pitch, a total of 25 hours. At the conclusion of the test run, a leakage test shall be performed in accordance with 4.7.4.2. Then the store shall be thoroughly inspected and no damage shall be evident. After inspection the store shall be completely filled with the water solution and vibrated, but not sloshed, for 10 minutes. At the conclusion of the test run another leakage test shall be performed in accordance with 4.7.4.2. Again the store shall be thoroughly inspected and no damage shall be evident.

4.7.6 Accelerated Load Tests: The purpose of these tests is to demonstrate that the store withstands aircraft acceleration conditions.

4.7.6.1 Test Specimen: The store shall be a complete production article in all respects.

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4.7.6.2 Test Setup: The store shall be mounted in a test fixture which supports the store on a suspension arrangement simulating the Aero 7A rack. Flight load tests may be conducted by either static or dynamic load test methods. Arrested landing and catapult loads shall be applied to the store dynamically.

4.7.6.3 Test Procedures: The store shall be tested for each test condition of Table I. Loads shall be applied and results recorded for approximately 20 percent increments up to design limit load and at 115 percent design limit load. The store shall be thoroughly inspected after testing for each condition. No damage or deformation shall be evident. After all the tests are carried to 115 percent design limit load, leakage and proof pressure tests shall be performed in accordance with 4.7.4.2 and 4.7.4.3. Then the store shall be tested to 150 percent of the design limit catapult load followed by inspection. The last accelerated load test shall be at 150 percent of flight test condition 2 in Table I. At the conclusion of all of the accelerated load tests a leakage test shall be performed in accordance with 4.7.4.2. Finally, the store shall be thoroughly inspected. There shall be no deformation which would adversely affect the mechanical operation of the store and no structural failure such as severely deformed, cracked, loose or broken parts or members.

4.7.7 Hose Load Test: The purpose of this test is to demonstrate that the hose reel and level wind guide withstand the loads imposed by the hose.

4.7.7.1 Test Specimen: The store shall be a complete production article from the center-section aft except that the tail cone shall be removed.

4.7.7.2 Test Setup: The store shall be mounted in a test fixture which supports the store on a suspension arrangement simulating the Aero 7A rack. The hose shall be at the full trail position and the hose reel shall be locked at the gear box. The hose shall be connected to a hydraulic jack or equivalent loading device. The arrangement shall be such that the hose load is applied in the horizontal plane at an angle of 18 1/2 degrees from the center of the level wind guide and in the direction to force the level wind guide toward the center of the store.

4.7.7.3 Test Procedure: Loads shall be applied and results recorded for 20 percent increments up to the 100 percent design limit load, 2000 pounds, and then to 115 percent load, 2300 pounds. After testing to 115 percent load the load shall be released and the reel and supporting structure shall be examined. No damage or deformation shall be evident. Then loads shall be applied and results recorded for 130, 140 and 150 percent of the 2000 pound design limit load. While the loads are applied the store shall be inspected and no deformation which would adversely affect the mechanical operation of the hose reel shall be evident. After testing to 150 percent load the load shall be released and the store shall be inspected to determine that no damage which would adversely affect the operation of the store has occurred. There shall be no structural failure such as severely deformed, cracked, loose or broken parts or members.

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TABLE I

Test Condition	Design Limit Load Factor Combination	Pitching Accel (Rad/Sec/Sec)	Yawing Accel (Rad/Sec/Sec)
1. Flight Store full	Nz = 7.5 (down) Ny = 1 (side) Nx = ± 2 (fwd & aft)	± 5	0
2. Flight Store full	Nz = 5 (down) Ny = 3 (side) Nx = ± 2 (fwd & aft)	± 5	0
3. Flight Store empty	Nz = 10 (down) Ny = 1.5 (side) Nx = ± 2 (fwd & aft)	± 6	0
4. Arrested Landing Store 1/4 full	Nz = 3 (down) Ny = 1.5 (side) Nx = 10 (fwd)	± 12	± 6
5. Catapult Store full	Nz = 3 (down) Ny = 1.5 (side) Nx = -9 (aft)	± 12	± 4

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4.7.8 Hose Reel Response Test: The purpose of this test is to demonstrate that the store is suitable for engagement by receiver aircraft at speeds from 2 to 8 feet per second (fps).

4.7.8.1 Test Specimen: The store shall be a complete production article.

4.7.8.2 Test Setup: The store shall be mounted in a test fixture which supports the store on a suspension arrangement simulating the Aero 7A rack. The store may be operated by an external source of hydraulic power. A means of applying 400 pounds of force on the extended drogue shall be provided to simulate drag load. A dummy air refueling probe fitted with a type MA-2 nozzle shall be employed. The probe shall be mounted on a travelling carriage or on a vehicle. The travelling probe arrangement shall be such that after engagement of the probe and drogue the forward motion of the probe decelerates uniformly to a position approximately 10 feet forward and stops. The store, extended hose, drogue, probe and direction of probe motion shall be in line. Motion picture coverage with 16mm equipment shall be provided.

4.7.8.3 Test Procedure: With the store operating in the EXTEND condition and the hose at full trail the drogue shall be engaged by the probe at speeds of 2, 4, 6, 8, 10 and 12 feet per second. At engagement speeds of 2, 4, 6 and 8 feet per second the response action shall not interfere with complete engagement of the nozzle and reception coupling and the hose shall not whip, loop, oscillate or develop excessive slack. Tests at speeds of 10 and 12 feet per second will be conducted for purposes of information. Upon completion of the tests the store shall be inspected and no damage to any part of the store shall be evident.

4.7.9 Store Endurance Test: The purpose of this test is to demonstrate that the store withstands 500 cycles of operation.

4.7.9.1 Test Specimen: The store shall be a complete production article.

4.7.9.2 Test Setup: The store shall be mounted in the production test wind tunnel used for acceptance tests and shall be powered by its ram-air turbine. The simulated drag load applied to the hose shall be 650 pounds for this test. Temperature shall be normal ambient temperature.

4.7.9.3 Test Procedure: The store shall be subjected to 500 cycles of operation. Each cycle shall consist of hose extension to the fully extended position and return to retracted and locked position. No adjustments, servicing of the store or replacement of parts shall be required during the test. Upon completion of the endurance test the store shall be subjected to and shall pass a ground operation test, 4.7.3. Finally, the store shall be thoroughly inspected. No damage except wear of the fuel hose shall be evident.

4.7.10 Hose Cutter Test: The purpose of this test is to demonstrate that the hose cutter functions satisfactorily in extreme temperature environment and as part of the store system.

4.7.10.1 Test Specimen: The store and the hose cutter shall be complete production articles.

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4.7.10.2 Test Procedure:

4.7.10.2.1 High Temperature: The hose cutter shall be mounted in a test chamber which contains an explosive fuel mixture. A short length of hose filled with JP-4 fuel and subjected to 300 pounds tension load shall pass through the cutter. The environmental temperature in the test chamber shall be raised to 150 degrees F and held for two hours. At the conclusion of the two hour soak period the cutter shall be fired. The cutter shall sever the hose and crimp and hold the upstream severed end of the hose. Ignition of the explosive fuel mixture shall not occur. Upon completion of the hose cutter firing the explosive fuel mixture in the chamber shall be ignited.

4.7.10.2.2 Low Temperature: The hose cutter shall be mounted in a test chamber. A short length of hose filled with JP-4 fuel and subject to 300 pounds tension load shall pass through the cutter. The environmental temperature in the test chamber shall be lowered to minus 65 degrees F and held for two hours. At the conclusion of the two hour soak period the cutter shall be fired. The cutter shall sever the hose and crimp and hold the upstream severed end of the hose.

4.7.10.2.3 Store Test: A complete store with hose cutter shall be mounted in the production test wind tunnel used for acceptance tests and shall be powered by its ram-air turbine. The hose shall be extended and 300 pounds simulated load drag shall be applied. The hose shall be full of fuel. With the ram-air turbine running the hose cutter shall be actuated by the hose jettison switch. The cutter shall sever the hose and crimp and hold the end remaining in the store. The store system shall be automatically shut down. The entire operation shall occur with safety. No fuel ignition shall occur.

4.7.11 Naval Service Suitability: The purpose of this test is to determine if the store conforms to the requirements of this specification and if it is suitable for naval use.

4.7.11.1 Test Specimen: The store shall be a complete production article.

4.7.11.2 Test Procedure: The service suitability evaluation will be conducted by a naval activity in accordance with instructions which will be issued by the Bureau of Naval Weapons. The evaluation will include flight tests to demonstrate compliance with the performance requirements of Section 3, and maintenance investigation to determine compliance with interchangeability requirements, 3.2.2. The contractor shall provide field service support for the evaluation.

5. PREPARATION FOR DELIVERY:

5.1 Part I - Air Refueling Store and Console:

5.1.1 Preservation: The store shall be preserved internally in accordance with Douglas Aircraft Company, Inc. Process Standard DPS 2.484-1.

5.1.2 Packing: The air refueling store and console shall be delivered in a shipping container, Part II of this specification.

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5.2 PART III - Special Support Equipment:

5.2.1 Preservation and Packaging: Preservation and packaging shall be in accordance with Method II of Specification MIL-P-116.

5.2.2 Marking: All interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use: The air refueling store covered by this specification is intended for rapid conversion of combat aircraft to the air refueling tanker configuration.

6.2 Ordering Data: Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) The Part of the air refueling store system being procured (see 1.2)
(c) The specific special support equipment to be furnished under Part III and the quantity of each item. The shipping container should be ordered under Part II of this specification, not included under Part III.
(d) The quantity of each Part to be furnished . (One preproduction store used for structural tests will not be delivered for service use).

6.3 Information Required with Bids:

6.3.1 Exceptions: Each bid shall include a statement as to whether or not any exceptions are taken to any part of this specification or to the drawings, specifications and other documents invoked herein.

6.3.2 Engineering Data: Each bid shall include a drawing of the installation of the high level float switch required by 3.7.7.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government Procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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